



Data Glacier

Your Deep Learning Partner

Project: Bank Marketing Campaign

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Github link:

https://github.com/priyanjalipatel/Data_Glacier_Final_Project

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Background

- **Business Problem**

- ❖ ABC Bank wants to increase **term deposit subscriptions**.
- ❖ Marketing campaigns (phone calls) are **costly and time-consuming**.
- ❖ Goal → **Predict customers most likely to subscribe** so bank can:
 - Save resources 💰
 - Improve targeting 🎯

Objective:

ABC Bank wants to sell its term deposit product to customers and before launching the product they want to develop a model which helps them in understanding whether a particular customer will buy their product or not (based on customer's past interaction with bank or other Financial Institution).

Data Exploration

- Source: UCI Bank Marketing Dataset (~45,000 records).
- Target Variable: **y** (yes/no for subscription).
- Features:
 - Demographics (age, job, marital, education, etc.)
 - Contact info (communication type, campaign outcomes)
 - Historical data (**pdays**, **previous**, etc.).

Challenges

- Imbalanced dataset (~90% No, ~10% Yes).
- Some misleading features (**duration** not practical in real-time).
- Need interpretable + scalable models.

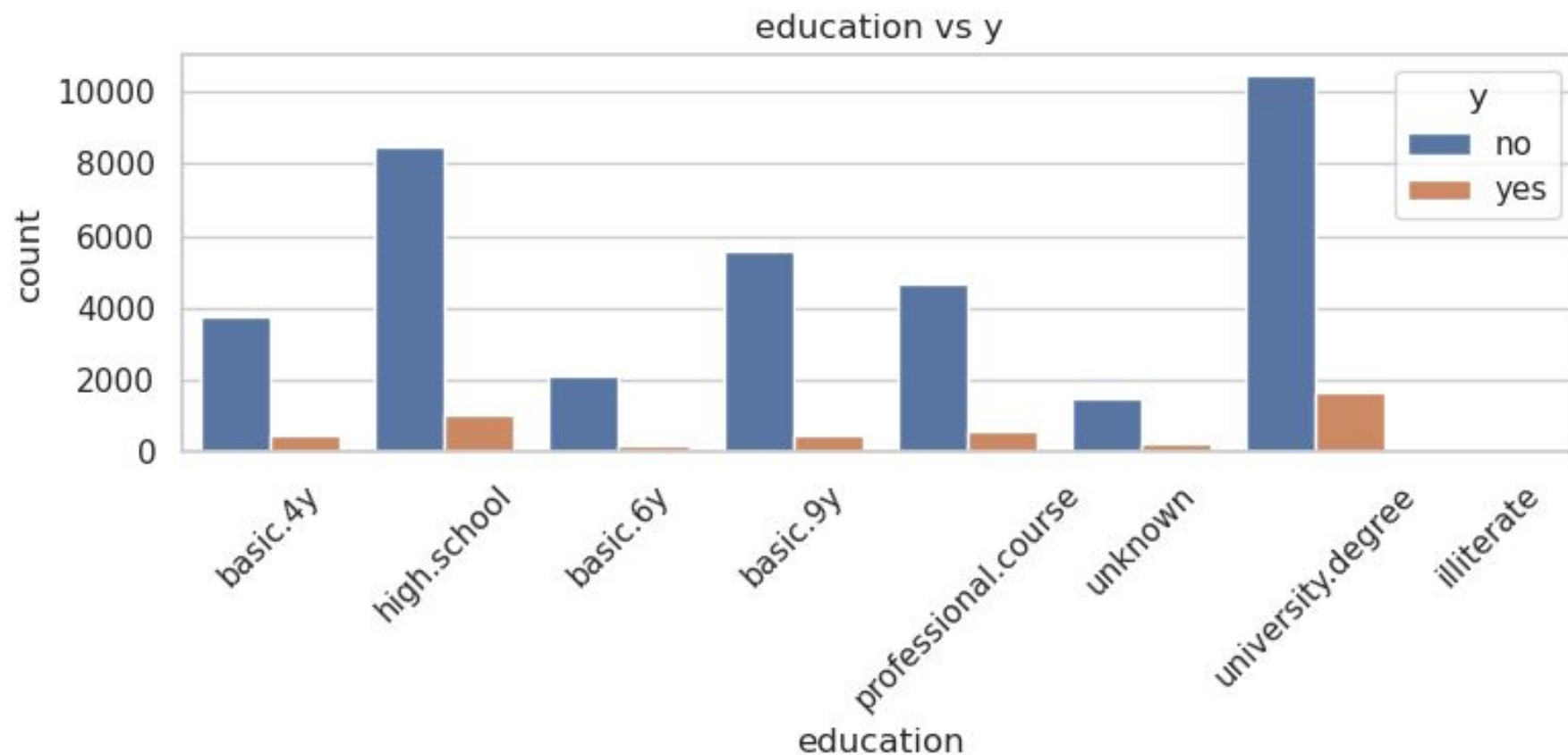
Approach

1. Business Understanding
2. Data Exploration (EDA)
3. Preprocessing (scaling, encoding, handling imbalance)
4. Model Building (Logistic Regression, Tree-based, Ensemble)

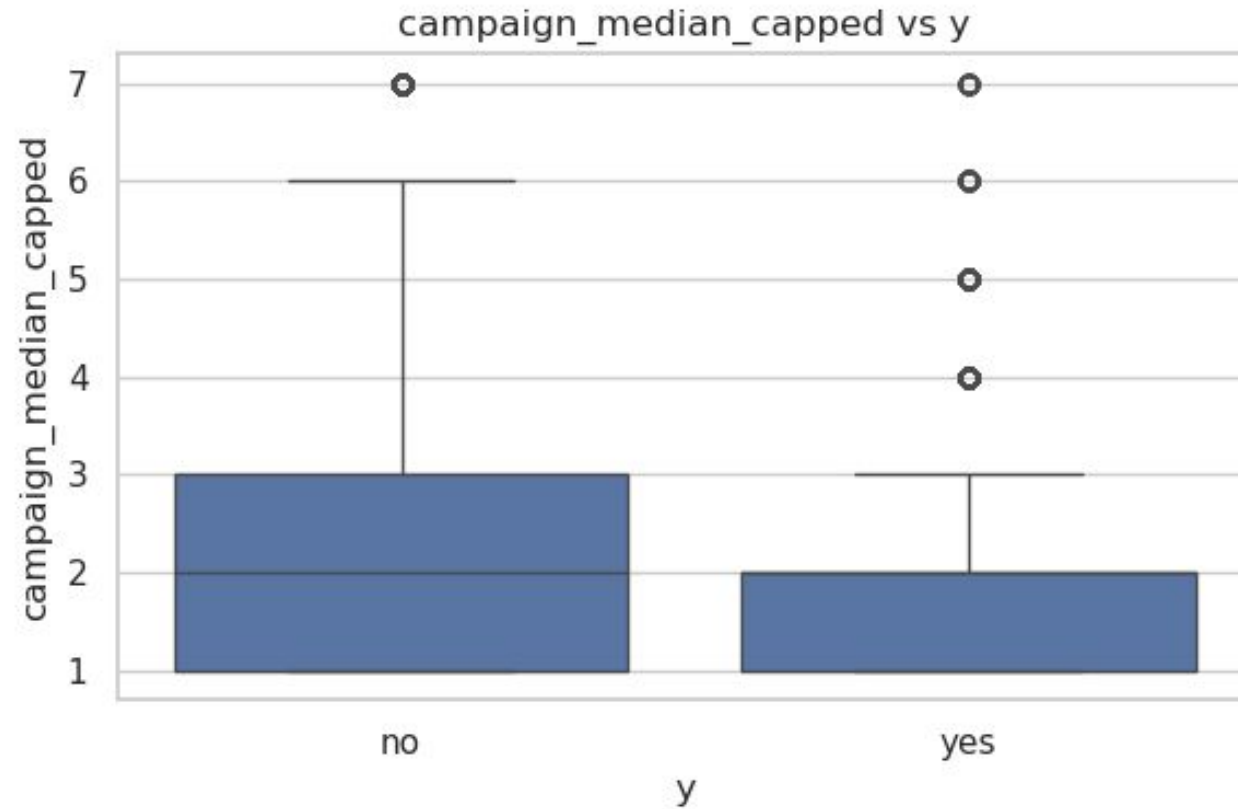
2. EDA Highlights

- Most customers contacted multiple times → low success rate.
- `pdays = 999` → indicates customer not contacted before (special case).
- Education & job type have clear influence on subscription.

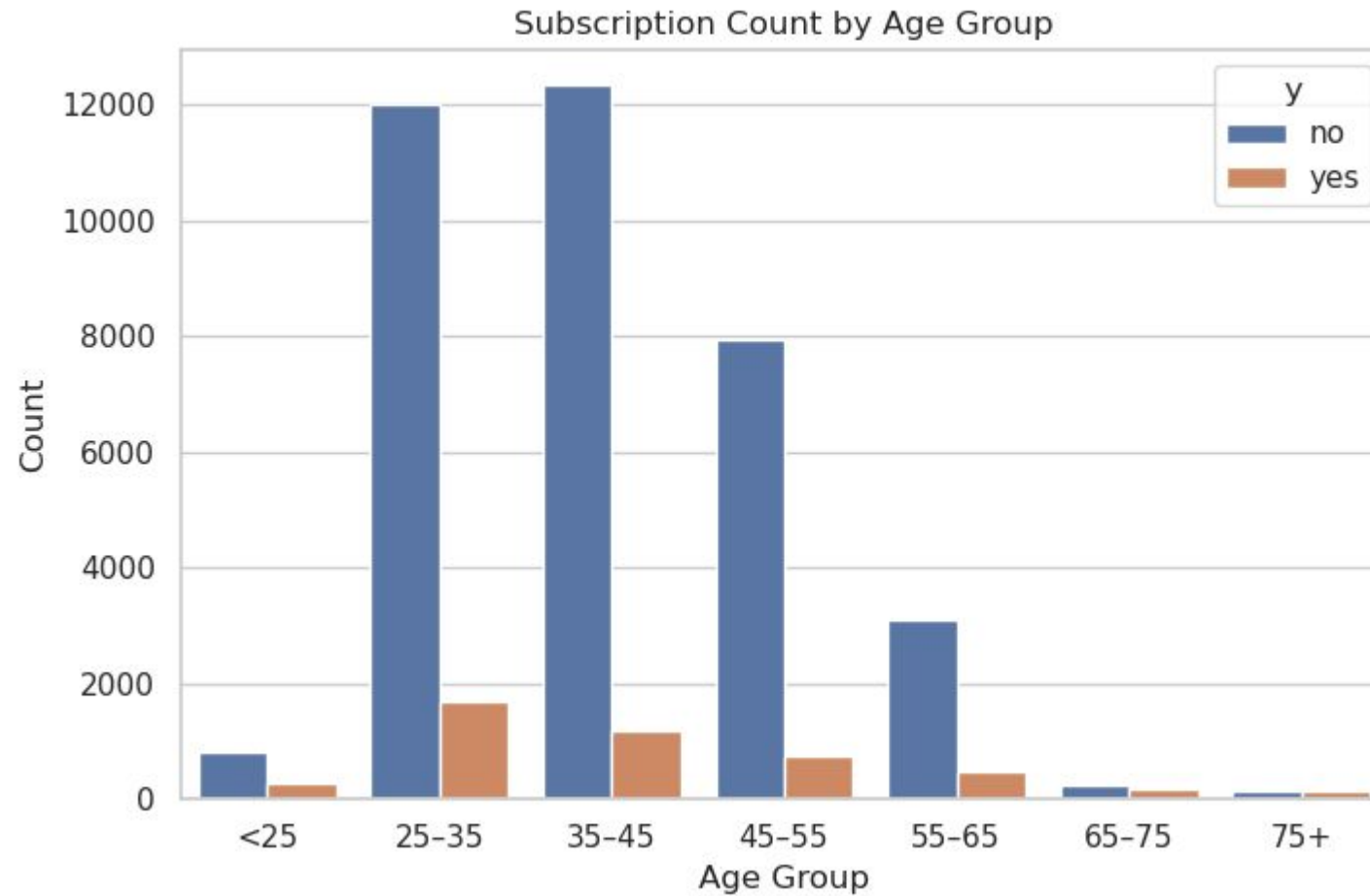
2. Exploratory Data Analysis (EDA)



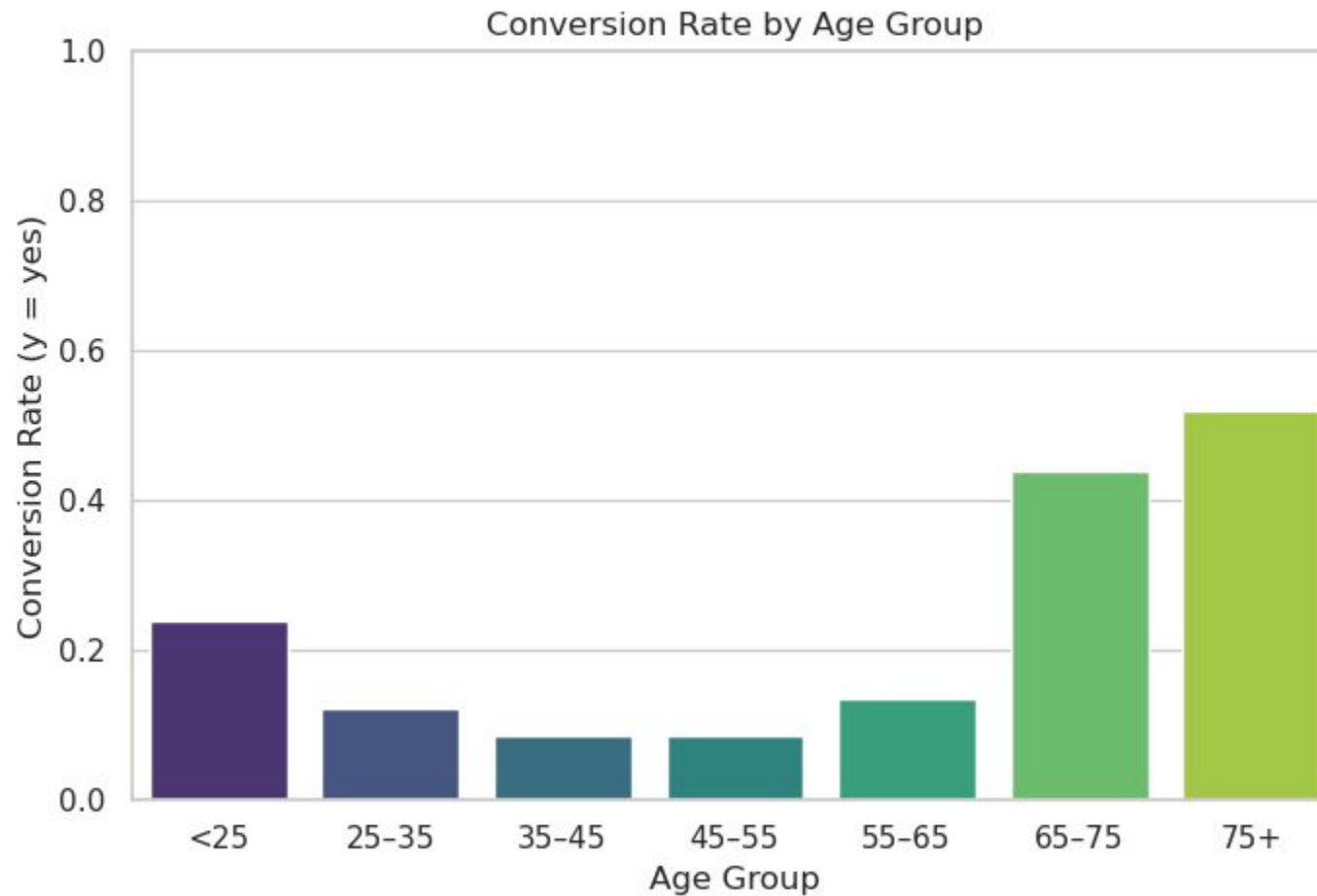
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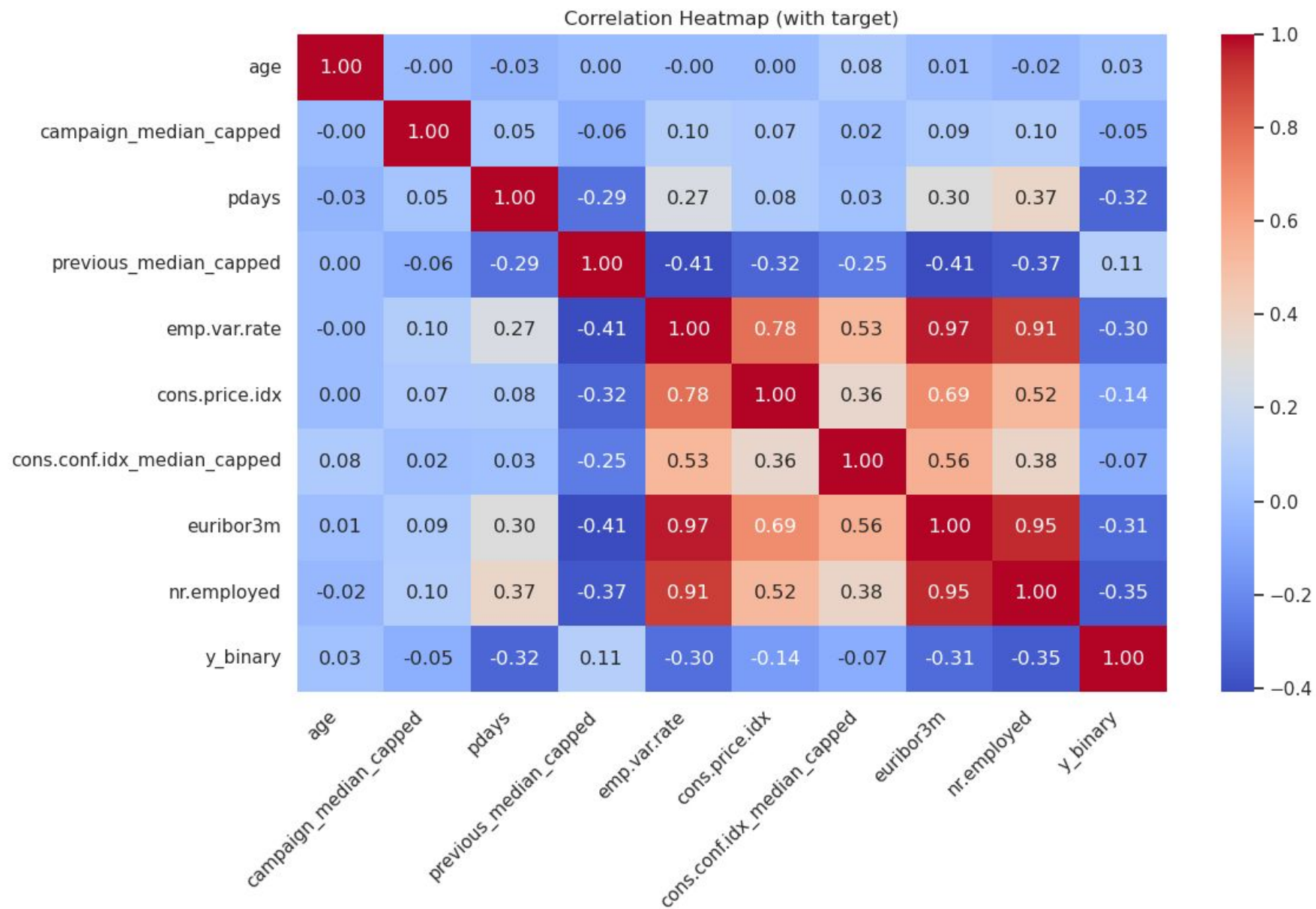
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Correlation heatmap



Correlation heatmap

Feature-wise Explanation

- *emp.var.rate (Employment Variation Rate)*

More employment variation → less likely to subscribe (economic uncertainty?)

- *euribor3m (Euro Interbank Offered Rate - 3 Months)*

Higher interest rates → less subscriptions (people may avoid locking into term deposits)

- *nr.employed (Number of Employees)*

High employment → possibly lower urgency to invest or switch products

Feature Engineering (Technical + Business-friendly wording)

- **Categorical Encoding:** Converted job, marital, education, etc. into numeric format (one-hot encoding)
- **Scaling:** Standardized continuous features (age, balance, duration, etc.) for fair comparison across models
- **Imbalance Handling:** Explored approaches like class weights (balanced) and oversampling (SMOTE)
- **Feature Selection:** Removed redundant/low-informative features

Models Compared

- Logistic Regression
- Decision Tree
- Naive Bayes
- Random Forest
- Gradient Boosting
- Evaluation Metrics: Accuracy, ROC-AUC, Recall, Precision

Final Slide (for Technical Users) → Recommended Models

Explored Families:

- **Linear Model** → Logistic Regression (baseline, interpretable).
- **Tree Model** → Decision Tree (simple but prone to overfitting).
- **Ensemble** → Random Forest (strong baseline, less overfitting).
- **Boosting** → Gradient Boosting (high performance, but less interpretable).
- **Others** → Naive Bayes (simple, but weaker performance).

Recommended Model:

- Logistic Regression can be chosen for **interpretability + strong performance** (important for banking regulations and business explanation).
- Ensemble models (Random Forest / Gradient Boosting) can be used as **supporting benchmarks**.
- **Business-Friendly (Interpretable):** Use **Decision Tree** for explainability.
- **Performance-Oriented:** Use **Gradient Boosting** for higher accuracy & recall.

Thank You



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